KPPSC PMS • GEOGRAPHY (PAPER-II)

EXECUTIVE MASTER EXAMINATION NOTES • Q.19: Geography of Development: Regional Disparities, Core-Periphery Dependency & Development Indicators

Target Examination	KPPSC PMS (Provincial Management Service) — Khyber Pakhtunkhwa
Paper & Subject	Geography Paper-II (Human, Economic & Regional Geography — 100 Marks)
EXAM QUESTIONS LINKED:	• 2025 Q.5: "Explain the geography of development and critically assess the factors responsible for regional disparities between developed and underdeveloped countries. How do these disparities influence global economic and political relations? (20 Marks)" • 2025 Q.8: "Explain the major economic and socio-demographic indicators used to assess national development. (20 Marks)" • 2018 Q.8(c): "Write short note on: Poverty alleviation strategies through spatial planning. (10 Marks)"
Syllabus Section	Section B & C: Economic & Regional Geography — Geography of development, regional disparities, core-periphery models
Allocated Marks & Time	20 Marks | 35 Minutes Exam Writing Allocation
Core Analytical Thesis	Economic development across geographic space is inherently uneven, producing polarized spatial structures characterized by dynamic, capital-accumulating 'Cores' and subordinate, resource-drained 'Peripheries'. Delineated theoretically through Gunnar Myrdal's Cumulative Causation (Backwash vs. Spread effects), Albert Hirschman's Polarization model, and John Friedmann's Core-Periphery paradigm, development cannot be measured solely by macroeconomic GDP, but requires multidimensional spatial indices (Human Development Index - HDI, Multidimensional Poverty Index - MPI). In Pakistan, this dynamic manifests in acute provincial and sub-regional disparities: the central and northern Punjab canal heartland functions as an overdeveloped core, while peripheral regions (Balochistan, southern Khyber Pakhtunkhwa, rural Sindh) suffer structural extraction, underinvestment, and severe human capital deficits.

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📌 EXAM ESSENTIAL TAKEAWAY

DEVELOPMENT INDICATORS METAMORPHOSIS: Shifted from narrow monetary metrics (Gross Domestic Product - GDP per capita) to holistic human capability frameworks: (1) Human Development Index (HDI: life expectancy, mean/expected years of schooling, GNI per capita); (2) Multidimensional Poverty Index (MPI: health, education, living standards deprivations); (3) Gender Inequality Index (GII).

GUNNAR MYRDAL'S CUMULATIVE CAUSATION (1957): Free market forces exacerbate spatial inequality through 'Backwash Effects' (capital, skilled labor, and raw materials drain from periphery to core); counteracted only when strong state intervention fosters 'Spread Effects' (diffusion of capital, technology, and market demand outward).

JOHN FRIEDMANN'S 4-STAGE CORE-PERIPHERY MODEL (1966): (1) Pre-Industrial (independent local centers, zero integration); (2) Core-Periphery (single dominant primate core draining periphery); (3) Dispersion

> (sub-cores emerge due to diseconomies at main core); (4) Spatial Integration (fully integrated, interconnected national polycentric system).
>
> PAKISTAN'S SPATIAL ASYMMETRY: High-HDI core enclave in central/northern Punjab (Lahore, Rawalpindi, Sialkot HDI > 0.70) contrasts sharply with acute multidimensional poverty in peripheral Balochistan (Awaran, Dera Bugti, Kohlu MPI > 70-80%) and merged tribal districts of KP, threatening national socio-political cohesion.

1. CONCEPTUAL MIND MAP & THEMATIC ROADMAP

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THE GEOGRAPHY OF UNEVEN DEVELOPMENT: THEORETICAL & REGIONAL MATRIX
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[1. EVOLUTION OF DEVELOPMENT METRICS]
  ├── Orthodox Growth Metric: GDP per Capita (Ignores spatial distribution, health, literacy, environmental costs).
  ├── Amartya Sen's Capability Approach: Development as Freedom ──► Expansion of substantive human choices.
  ├── Human Development Index (HDI): Geometric mean of Longevity (Life Expectancy) + Education + Income (GNI).
  └── Multidimensional Poverty Index (MPI): Captures simultaneous overlapping deprivations at the household level.
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        ▼ (The Structural Mechanics of Uneven Spatial Growth)
[2. CORE THEORETICAL PARADIGMS OF SPATIAL POLARIZATION]
  ├── GUNNAR MYRDAL (1957): "Theory of Circular and Cumulative Causation"
  │     ├── The Core Spark: Initial geographical advantage (port, mineral, state capital) triggers inward investment.
  │     ├── The Backwash Mechanism: Periphery loses best human talent (Brain Drain) + capital to high-yield core.
  │     └── The Spread Effect: Core's economic momentum eventually spills outward (buying peripheral food/minerals).
  │           Crucial Rule: In free market capitalism, BACKWASH ALWAYS OVERWHELMS SPREAD unless state intervenes!
  ├── ALBERT HIRSCHMAN (1958): "Unbalanced Growth: Polarization vs. Trickle-Down"
  │     └── Polarization effects (centripetal forces) dominate early development; requires deliberate state planning.
  └── JOHN FRIEDMANN (1966): "Four-Stage Spatial Evolution Model"
        Stage 1 (Pre-Industrial) ──► Stage 2 (Core-Periphery) ──► Stage 3 (Sub-Core Dispersion) ──► Stage 4 (Spatial
Integration).
        │
        ▼ (The Pakistan National Disparities Landscape)
[3. THE PROVINCIAL SPATIAL CHASM IN PAKISTAN]
  ├── THE INDUSTRIAL & AGRARIAN CORE (Central/North Punjab & Karachi)
  │     ├── Features: Dense infrastructure, G.T. Road/M-2 corridor, high industrial density, low MPI (<15%).
  │     └── Representative Districts: Lahore, Islamabad, Rawalpindi, Faisalabad, Sialkot, Gujranwala.
  └── THE EXTRACTIVE & NEGLECTED PERIPHERY
        ├── Balochistan Periphery: 44% of national land, provides natural gas (Sui) and minerals (Reko Diq, Saindak),
        │     yet retains Pakistan's highest MPI (>70% in rural districts), lowest literacy (<20% female).
        ├── Khyber Pakhtunkhwa Dualism: Prosperous Peshawar/Mardan valley vs. Impoverished Southern & Merged Districts
(Waziristan).
        └── Rural Sindh Periphery: Extreme feudal landholding, high malnutrition (stunting >45%), decaying rural
towns.
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2. ANALYTICAL DIMENSIONS & SOCIO-SPATIAL/THEORETICAL MATRIX

Development Theory / Model	Core Spatial Mechanism & Driving Dynamics	Treatment of Market Forces vs. State Role	Empirical Application to Regional Planning
Myrdal's Cumulative Causation (1957)	Initial advantage triggers cumulative cycle. Backwash effects (capital/talent drain) overpower Spread effects.	Free market forces widen regional inequality; requires strong, targeted state redistributive intervention.	Explains why southern Punjab and Balochistan remain underdeveloped while Lahore/Karachi continuously accumulate capital.
Hirschman's Polarization & Trickle-Down (1958)	Growth does not appear everywhere simultaneously; it manifests at specific growth points (poles).	Polarization dominates early stages; trickle-down takes decades to manifest without government spatial incentives.	State builds infrastructure in backward growth poles (e.g., tax holidays in Hattar Industrial Estate, Gadoon Amazai in KP).
Friedmann's Core-Periphery Model (1966)	Four evolutionary stages: Pre-industrial -> Core-Periphery -> Semi-Periphery Dispersion -> Spatial Integration.	Dynamic spatial transition. Periphery is structurally dependent upon, and exploited by, the metropolitan core.	Pakistan is currently trapped in Stage 2 (dualistic Core-Periphery), with Karachi/Lahore dominating national wealth.

Development Theory / Model	Core Spatial Mechanism & Driving Dynamics	Treatment of Market Forces vs. State Role	Empirical Application to Regional Planning
Wallerstein's World Systems Theory (1974)	Global tripartite hierarchy: Core (high-tech/capital), Semi-Periphery (manufacturing), Periphery (raw materials).	Unequal exchange: Surplus value is systematically transferred from peripheral nations to core capitalist states.	Pakistan acts as a global periphery exporting raw textiles and importing high-tech capital goods, cementing trade deficits.
Multidimensional Poverty Index (MPI)	Measures 3 dimensions across 10 indicators: Education, Health, Living Standards (water, electricity, assets).	Transcends income poverty; identifies exact spatial deprivation pockets at district and union council levels.	Pakistan National MPI: Over 38% national multidimensional poverty, soaring to >70% in rural Balochistan and Merged Districts of KP.

3. 20-MARK MASTER MODEL ESSAY (KPPSC PMS BENCHMARK)

Introduction: The Spatial Unevenness of Global Development

Development is fundamentally a geographical phenomenon. Far from diffusing evenly across national landscapes, economic growth, capital accumulation, and human prosperity concentrate in specific, highly localized geographic nodes, leaving expansive surrounding hinterlands in states of relative deprivation and economic stagnation. In development geography, the analysis of spatial inequality has evolved from simplistic monetary comparisons of Gross Domestic Product (GDP) per capita toward sophisticated multidimensional human capability frameworks. Pioneered by Swedish Nobel laureate Gunnar Myrdal, Albert Hirschman, and John Friedmann, development theory demonstrates that spatial polarization is not a temporary market anomaly, but the natural, inherent tendency of capitalist market mechanisms. In Pakistan, severe inter-provincial and intra-provincial disparities represent the supreme structural threat to national integration, economic resilience, and social cohesion.

The Epistemological Evolution of Development Metrics: From GDP to HDI and MPI

Historically, the orthodox economic paradigm measured development strictly through macroeconomic monetary growth: Gross National Product (GNP) and Gross Domestic Product (GDP) per capita. However, geographers and development economists demonstrated that GDP is fundamentally flawed as a development metric: (1) it obscures extreme spatial inequality, treating a billionaire and an impoverished tenant farmer as a statistical average; (2) it counts environmental destruction (deforestation, oil spills) as positive economic output; and (3) it ignores unpaid domestic labor, human health, and political freedom. In 1990, visionary Pakistani economist Dr. Mahbub ul Haq, collaborating with Indian Nobel laureate Amartya Sen, fundamentally inverted this paradigm by creating the Human Development Index (HDI) under the United Nations Development Programme (UNDP). Sen's 'Capability Approach' asserted that development is the expansion of human freedom—enhancing people's capability to lead long, healthy, creative, and educated lives. The HDI constructs a holistic, composite geometric index across three equally weighted dimensions: 1. Health Dimension: Measured by Life Expectancy at Birth (reflecting infant survival, nutrition, and healthcare access). 2. Education Dimension: Measured by Mean Years of Schooling for adults and Expected Years of Schooling for children. 3. Living Standards Dimension: Measured by Gross National Income (GNI) per capita adjusted for Purchasing Power Parity (PPP). To capture overlapping deprivations at the granular household level, the Oxford Poverty and Human Development Initiative (OPHI) and UNDP formulated the Multidimensional Poverty Index (MPI). Rather than relying on an arbitrary income line ($2.15/day), the MPI measures ten specific indicators across health (nutrition, child mortality), education (years of schooling, school attendance), and living standards (cooking fuel, sanitation, drinking water, electricity, housing, and assets). An individual is multidimensionally poor if they are deprived in at least one-third of the weighted indicators, providing geographers with an extraordinary spatial lens to map human deprivation.

Theoretical Paradigms of Spatial Polarization: Myrdal, Hirschman, and Friedmann

Geographers utilize three seminal theoretical models to explain why regional disparities emerge and perpetuate: 1. Gunnar Myrdal's Cumulative Causation (1957): Myrdal rejected the neoclassical economic assumption that free markets automatically restore regional equilibrium. Instead, he formulated the principle of 'Circular and Cumulative Causation'. Once a specific geographic region gains an initial historical advantage (such as a deep-water port, colonial

administrative investment, or mineral deposits), it triggers a self-reinforcing upward spiral: capital flows inward, infrastructure expands, and skilled labor migrates to the growing center. This dynamic sets in motion two opposing spatial forces: - Backwash Effects: The growing economic core drains the surrounding peripheral regions of their most productive resources. The periphery suffers a catastrophic loss of young, educated labor ('brain drain'), local banking capital is funneled to high-yield core investments, and cheap manufactured goods from the core wipe out traditional rural handicraft industries. - Spread Effects: The core's economic vitality eventually spills outward into the periphery, as the core purchases peripheral agricultural produce and raw materials, and as rising congestion in the core encourages industrialists to establish branch plants in outer regions. Crucially, Myrdal proved that under unregulated capitalist market conditions, Backwash Effects always overpower Spread Effects, causing regional disparities to widen inexorably unless aggressive, targeted state intervention redistributes capital and infrastructure to peripheral zones. 2. Albert Hirschman's Unbalanced Growth (1958): Working independently in Latin America, Hirschman reached identical conclusions, terming these dual forces 'Polarization Effects' (centripetal forces concentrating capital in growth poles) and 'Trickle-Down Effects' (centrifugal diffusion). Hirschman argued that economic growth must be deliberately unbalanced, but warned that governments must actively manage the timing of interventions before polarization tears the social fabric of the state apart. 3. John Friedmann's Core-Periphery Evolutionary Model (1966): Friedmann synthesized spatial development into four distinct historical stages: (1) Stage 1 (Pre-Industrial): Independent local settlement centers with minimal inter-regional trade; (2) Stage 2 (Core-Periphery / Incipient Industrialization): A single massive primate city core dominates the nation, ruthlessly exploiting an expansive, impoverished periphery (a stage typified by many post-colonial developing nations); (3) Stage 3 (Dispersion / Maturing Industrialization): Severe diseconomies of scale at the core (traffic congestion, exorbitant real estate, pollution) combined with state decentralization policies stimulate the rise of secondary growth poles (sub-cores) across the periphery; (4) Stage 4 (Spatial Integration): A fully integrated, polycentric national urban hierarchy emerges, connected by advanced transport and digital grids, achieving regional equilibrium.

The Pakistan Landscape: The Chasm Between Core and Periphery

Pakistan represents a textbook empirical manifestation of a nation trapped in Friedmann's Stage 2 Core-Periphery distortion, where massive regional disparities threaten national federation: - The Developed Core (Central and Northern Punjab): Centered upon the fertile, canal-irrigated plains of central and northern Punjab, stretching along the historic Grand Trunk Road and motorways from Lahore through Gujranwala, Sialkot, Faisalabad, to Rawalpindi-Islamabad. This core houses over 50% of the national population, monopolizes more than 65% of the national industrial and manufacturing capacity, and captures the lion's share of federal development spending. Districts like Lahore, Rawalpindi, and Sialkot exhibit high human development indicators (HDI > 0.70 to 0.75, multidimensional poverty rates below 10%), featuring world-class tertiary hospitals, expressways, fiber-optic connectivity, and premier universities. - The Deep Periphery (Balochistan): Covering nearly 44% of Pakistan's total land area, Balochistan functions as an archetypal exploited periphery. Despite supplying the national core with natural gas from the historic Sui field since 1952, and possessing multi-billion-dollar mineral deposits (Reko Diq copper-gold, Saindak), Balochistan retains Pakistan's lowest developmental indicators. Over 70% of the rural population lives in acute multidimensional poverty (soaring to over 85% in districts like Dera Bugti, Kohlu, and Awaran). Basic human infrastructure is virtually absent: female literacy languishes below 20%, infant mortality is the highest in the country, and thousands of villages lack electricity, paved access roads, or clean drinking water. This stark core-periphery exploitation fuels persistent nationalist resentment and political alienation. - Regional Dualism in Khyber Pakhtunkhwa: KP exhibits intense internal polarization between its central valley core (the fertile, urbanized Peshawar-Mardan-Abbottabad triangle, exhibiting reasonable industrial and educational facilities) and its deeply marginalized southern and western periphery (the newly Merged Tribal Districts of former FATA: North and South Waziristan, Kurram, Khyber). Devastated by decades of geopolitical conflict, displacement, and total absence of municipal infrastructure, the merged districts exhibit multidimensional poverty rates exceeding 70%. - The Agrarian Feudal Periphery of Rural Sindh: While the megacity of Karachi functions as a commercial financial island, rural Sindh (districts like Tharparkar, Badin, Jacobabad, Thatta) represents a severely neglected agrarian periphery. Paralyzed by entrenched feudal tenancy structures, rural Sindh exhibits shocking child stunting rates exceeding 45%, collapsing public school infrastructure, and widespread waterborne disease epidemics.

Mechanisms Perpetuating Spatial Disparity: Political Economy and Fiscal Centralization

Why do regional disparities persist in Pakistan? Spatial inequalities are institutionalized through several structural mechanisms: 1. Fiscal Allocation Distortions: Historically, resources under the National Finance Commission (NFC) Award were distributed solely on the basis of population size. Because Punjab contained more than half the national population, it captured the overwhelming majority of federal revenue. While the 7th NFC Award (2009) introduced multiple criteria (poverty/backwardness, revenue collection, inverse population density), population still commands an 82% weighting, continuing to starve vast, sparsely populated peripheries like Balochistan of capital for basic infrastructure. 2. Inward Investment Bias: Both public infrastructure investments (metro buses, motorways, signal-free corridors) and private commercial capital are concentrated in the developed core, where purchasing power and industrial agglomeration ensure higher immediate financial returns, reinforcing Myrdal's backwash drain. 3. The Feudal Rent-Seeking Complex: In peripheral rural Sindh and Balochistan, traditional feudal and tribal elites (waderas, sardars) have historically co-opted state funds, actively resisting mass public education and land reforms to preserve their hereditary political dominance over impoverished populations.

Conclusion: Strategic Imperatives for Balanced Polycentric Spatial Planning

In conclusion, the geography of development demonstrates that spatial inequality is not merely an unfortunate byproduct of economic growth, but an active barrier that strangles national developmental potential. A federation cannot survive indefinitely when its core prospers while its geographical peripheries starve. To achieve balanced, sustainable national integration, Pakistan must execute a comprehensive spatial redistribution strategy: (1) fundamentally revise the National Finance Commission (NFC) allocation formula, elevating 'inverse population density' and 'multidimensional poverty' to at least 40% of the criteria; (2) utilize Special Economic Zones (SEZs) under CPEC specifically to seed growth poles in peripheral regions (such as the Bostan SEZ in Balochistan and Rashakai SEZ in KP); (3) implement targeted, ring-fenced federal human capital investments in maternal health, clean drinking water, and technical vocational institutes across Balochistan and the merged tribal districts; and (4) construct inter-regional lateral transport corridors (such as the CPEC Western Alignment) to break the monopoly of the central motorway corridor, integrating remote peripheral communities into the national economic bloodstream.

4. SCHOLARLY PERSPECTIVES & GEOGRAPHICAL AXIOMS TABLE

Scholar / Geographer	School of Thought / Paradigm	Quotable Philosophical Thesis / Axiom
Gunnar Myrdal (1957)	Theory of Cumulative Causation	*"Play of the forces in the market normally tends to increase, rather than to decrease, the inequalities between regions... The backwash effects continuously draw wealth from the periphery to the core."*
Dr. Mahbub ul Haq (1990)	Creator of the Human Development Index (HDI)	*"The real wealth of a nation is its people. The basic objective of development is to create an enabling environment for people to enjoy long, healthy, and creative lives."*
Amartya Sen (1999)	Development as Freedom	*"Development cannot be judged merely by the accumulation of commodities or the growth of gross national product; it must be evaluated by the substantial freedoms that people have to lead the lives they value."*
John Friedmann (1966)	Regional Development Policy: A Case Study	*"The core-periphery relationship is essentially a colonial relationship: the core dominates, extracts, and accumulates, while the periphery remains in a state of structural dependency."*
Akhtar Hameed Khan (1985)	Rural Development Pioneer	*"Our development planning is fatally top-heavy; we build grand monuments in the capital while the peripheral villages rot in medieval neglect."*